Position comparison excluding anomalous records

Independent unwrap per position, then one constant whole-curve 360-deg shift at the first plotted frequency; no smoothing.
Gray dashed: anomalous record. Thin dotted: nonconverged/provisional fit. Curve presence does not establish physical modal identity.

z = -0.5 mm z = 0 mm z = 0.5 mm z = 1 mm

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)